Naimiris A Profit-Driven Smart Farming Model for Scotch Bonnet Peppers

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Project Proposal Report

Savinda R.M.D – IT22124630

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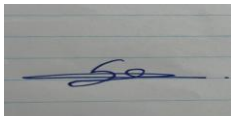
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	**Student ID**	**Signature**
Savinda R.M.D	**IT22124630**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Signature	**Signature**	**Signature**

ABSTRACT

The global agricultural industry is in the midst of a paradigm shift with the pressing need to optimize crop yields in the face of climate volatility, increasing scarcity of resources, and the overall rise in cultivation costs. In the case of Sri Lanka, the financial implications of inefficient agricultural practices are significant, as exemplified by the annual expenditure of 100 million USD on importing chilies to bridge the gap between local production and national demand. In this context, the cultivation of Capsicum chinense, popularly known as Nai Miris or Scotch Bonnet pepper, is an extremely profitable agricultural venture owing to its pungency, demand, and inherent drought tolerance. However, small-scale farmers are often subject to significant crop losses owing to their inability to effectively monitor micro-climatic conditions and correlate them with the weekly phenotypic progress of the crop. Currently, the smart agriculture landscape is highly fragmented, with most solutions failing to isolate Internet of Things (IoT) environmental monitoring and Artificial Intelligence (AI) based disease detection, and most importantly, crop growth progression analysis.

The proposal of this study is the creation of an exhaustive technology-aided A-to-Z farming guidance system that is specifically designed and developed for the early-stage cultivation of Nai Miris. This solution is a pioneering technology that uses a unique data fusion technique that fuses real-time IoT environmental sensing with Edge AI computer vision in assessing the growth of the crop throughout a specified four-week early growth lifecycle. By using an ESP32 technology-based sensor network that carefully monitors the soil moisture, EC, temperature, and humidity levels in the environment, as well as using a YOLOv8 and ResNet-18 technology-based computer vision module that analyzes images of the crop on a weekly basis, the solution is able to determine whether the crop is growing normally, slowly, or abnormally. A unique data fusion algorithm is used in correlating abnormal growth patterns with real-time environmental sensing in generating highly localized and specific farming guidance.

The envisioned outcome of this research is an affordable and highly scalable decision support framework that would transform farming from an essentially reactive and visually inspected activity into a proactive and data-driven science. This approach enables farmers to detect delays in crop development and micro-environmental issues far more rapidly than conventional methods, thus enhancing decision support and hence crop productivity. From an economic perspective, this system bridges an essential void in the Sri Lanka Agri-Tech space, hence supporting national macroeconomic strategies that promote import substitution while also being aligned with international SDGs for poverty reduction, eradication of hunger, and climate action.8Keywords: Smart Agriculture, Edge AI, Internet of Things, Precision Farming, Crop Phenotyping, Capsicum Chinen’s

ACKNOWLEDGEMENT

The global agricultural sector is in the midst of a paradigm shift due to the pressing need to maximize crop yield in the face of rising climate change risks, increased scarcity of resources, and the rising costs of cultivation. The economic impact of inefficient agricultural production in Sri Lanka is enormous, as exemplified by the country's annual expenditure of USD 100 million on the import of chilies to bridge the gap between local production and national requirements. In this context, the cultivation of Capsicum chinense, also known as Nai Miris or Scotch Bonnet pepper, is seen as a highly profitable agricultural venture owing to its extreme pungency, high market requirements, and natural drought resistance.4 However, small-scale farmers face high crop yield loss due to the lack of the ability to monitor the micro-climatic conditions and relate them to the weekly phenotypic development of the crop. The existing smart agriculture technology solutions are highly fragmented in nature, with the system failing to integrate the monitoring of the Internet of Things (IoT) with Artificial Intelligence (AI) technology-based disease detection and also failing to provide crop development progression analysis.

This research proposes the development of an exhaustive technology-assisted A to Z farming guidance system specifically designed and engineered for the early stage cultivation of the crop variety known as Nai Miris. The proposed solution will pioneer a novel data fusion technique by synergistically combining real-time IoT environmental sensing with Edge AI computer vision to continuously evaluate the development of the crop over a specified four-week early growth lifecycle. The proposed solution will utilize an ESP32-based IoT sensor network to meticulously monitor the environmental conditions of soil moisture, electrical conductivity (EC), ambient temperature, and humidity, as well as a YOLOv8 and ResNet-18-based AI-powered computer vision module to analyze plant images once per week to evaluate whether the crop development is occurring normally, slowly, or abnormally under physiological stress. The proposed solution will utilize a novel data fusion algorithm to correlate crop development anomalies with real-time environmental conditions to provide highly localized and actionable farming guidance.

The envisioned outcome of this research is the development of a cost-effective and highly scalable decision support system that transforms the farming business from a reactive and visually inspected task to a proactive and data-based science. This integrated system enables the identification of crop development delays and micro-environmental issues much before the onset of such problems using conventional techniques, thus improving the decision-making process and optimizing crop productivity. From the business perspective, this system fills the existing gap in the Sri Lankan Agri-Tech market and aligns with the country’s national macroeconomic goals by reducing import dependency while also meeting the global SDGs related to poverty reduction, zero hunger, and climate change mitigation.

Keywords: Smart Agriculture, Edge AI, Internet of Things, Precision Farming, Crop Phenotyping, Capsicum chinense

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
CNN	Convolutional Neural Network
YOLO	You Only Look Once
EC	Electrical Conductivity
DOA	Department of Agriculture
FAO	Food and Agriculture Organization
ADB	Asian Development Bank
IPCC	Intergovernmental Panel on Climate Change
IWMI	International Water Management Institute
SDG	Sustainable Development Goal
HCI	Human-Computer Interaction
AIMS	Autonomous Intelligent Machines and Systems
MQTT	Message Queuing Telemetry Transport
HTTP	Hypertext Transfer Protocol
LoRaWAN	Long Range Wide Area Network
RGB	Red, Green, Blue (Color Model)
mAP	Mean Average Precision
FLOPs	Floating Point Operations
ESP32	Espressif 32-bit Microcontroller
API	Application Programming Interface

Abbreviation	Full Form
GPS	Global Positioning System
UI	User Interface
UX	User Experience

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1. INTRODUCTION

1.1 Background and Context

The integration of digital technologies into agriculture, widely recognized as precision agriculture or smart farming, has emerged as a transformative approach to addressing global food security challenges in the 21st century. Rapid population growth, climate change, and increasing pressure on natural resources have intensified the need for efficient and sustainable agricultural practices**,** [1] [2]. Precision agriculture leverages modern technologies such as the Internet of Things (IoT), artificial intelligence (AI), machine learning, and data analytics to optimize farming operations, improve crop yield, and ensure efficient resource utilization [3]**,** [4].

Globally, agriculture is transitioning from traditional experience-based practices to data-driven decision-making systems. These systems enable farmers to monitor environmental conditions in real time, predict crop behavior, and apply inputs such as water and fertilizers more precisely [5]. As a result, smart farming has gained significant attention from researchers, governments, and industry stakeholders as a key solution for sustainable agricultural development.

In the context of Sri Lanka, agriculture remains a cornerstone of the national economy and rural livelihood. The sector supports more than 1.7 million smallholder farmers and contributes significantly to food security and employment [6]. However, Sri Lanka's agricultural sector faces numerous challenges, including unpredictable weather patterns, prolonged droughts, floods, soil degradation, and rising production costs [7]**,** [8]. Additionally, a major issue lies in the lack of efficient knowledge transfer between agricultural experts and farmers, leading to suboptimal farming practices and reduced productivity [9].

Among the various crops cultivated in Sri Lanka, chili pepper (Capsicum spp.) holds substantial economic and cultural importance. It is a staple ingredient in local cuisine and has high demand in both domestic and export markets. Despite favorable agro-climatic conditions, Sri Lanka experiences a significant production deficit in chili cultivation, resulting in heavy reliance on imports and increased vulnerability to global market fluctuations [10].

To address this issue, there has been increasing interest in cultivating high-value chili varieties such as Capsicum chinense, locally known as Nai Miris (Scotch Bonnet). This variety is known for its extremely high pungency, drought tolerance, and ability to produce yields throughout the year [11]**,** [12]. These characteristics make it a promisingcrop for improving local production and reducing import dependency.

Agronomic studies indicate that the successful cultivation of Nai Miris requires careful management of environmental parameters such as soil moisture, temperature, humidity, pH levels, and electrical conductivity (EC), which directly influence nutrient uptake and plant growth [13]. Maintaining optimal conditions is particularly important during the early growth stages, where plant development significantly impacts final yield.

However, smallholder farmers often lack access to tools and technologies that enable precise monitoring of these parameters. Traditional farming practices rely heavily on manual observation and experience-based decision-making, which can lead to delayed detection of crop stress and inefficient use of resources [14]. By the time visible symptoms such as leaf discoloration or wilting appear, the plant may have already suffered irreversible damage, resulting in reduced yield and financial loss [15].

To overcome these limitations, smart agriculture systems integrating IoT and AI technologies have been proposed. IoT sensors enable continuous monitoring of environmental conditions, while AI models analyze both sensor data and visual information to provide insights into crop health and development [16], [17]. These systems have the potential to transform agriculture into a proactive, data-driven process by providing real-time recommendations to farmers.

Despite these advancements, existing solutions often operate in isolation, focusing either on environmental monitoring or image-based analysis. There is a growing need for integrated systems that combine multiple data sources to provide comprehensive and actionable insights. This research aims to address this need by developing a technology-assisted A-to-Z farming guidance system that combines IoT-based environmental sensing with AI-driven growth stage analysis for Nai Miris cultivation.

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1.2 Literature Review

The rapid advancement of digital technologies has significantly influenced modern agricultural practices, leading to the development of smart farming systems that utilize IoT and AI for crop monitoring and management [18].

One of the key areas of research in smart agriculture is the application of deep learning techniques for plant analysis. Convolutional Neural Networks (CNNs) have been widely used for tasks such as plant disease detection, leaf classification, and yield prediction [19]. More recently, object detection models such as YOLO (You Only Look Once) have gained popularity due to their ability to perform real-time detection with high accuracy [20].

Studies have demonstrated the effectiveness of YOLO-based models in detecting plant growth stages and identifying crop maturity levels. For instance, recent research has shown that YOLO architectures can achieve high precision in classifying plant development stages under controlled conditions [21]. Additionally, improvements in lightweight deep learning models have enabled their deployment on edge devices with limited computational resources, making them suitable for real-world agricultural applications [22].

Parallel to advancements in AI, IoT-based systems have been extensively used for environmental monitoring in agriculture. These systems utilize sensors to collect real-time data on soil moisture, temperature, humidity, and electrical conductivity, which are critical factors for crop growth [23]. Such data enables farmers to implement precision irrigation and fertigation strategies, improving resource efficiency and crop productivity.

Furthermore, the integration of edge computing in smart agriculture has enhanced system performance by enabling local data processing and reducing dependence on cloud infrastructure [24]. Edge computing also improves system reliability in rural areas with limited internet connectivity.

Despite these technological advancements, existing systems exhibit several limitations. Many IoT-based solutions rely on predefined threshold values to trigger alerts, which may not accurately reflect the dynamic nature of plant growth and environmental interactions. Similarly,

AI-based systems often focus on detecting diseases or classifying crops into broad categories, without considering detailed growth progression.

In addition, commercial platforms such as GeoGoviya and similar smart farming applications primarily provide macro-level insights or reactive diagnostic capabilities [25]. These systems lack the ability to continuously monitor individual plants and provide stage-specific recommendations based on both environmental and visual data.

Table 1omparison Chart

VERY IMPORTANT: Integrated AI Growth Stage Monitoring + IoT Environmental Validation +Farming Guidance Comparison Chart

Task / Criteria	Research A - Disease Detection Systems	Research B - General Crop Growth Detection	Research C - Standalone IoT Monitoring	**Proposed System** (Nai Miris Growth & Env.)
Stage-Wise Weekly AI Growth Monitoring	✗	✓	✗	✓
Integrated IoT Environmental Validation	✗	✗	✓	✓
Growth-Stage Based Farming Guidance	✗	✗	✗	✓
Low-Cost System for Small-Scale Farmers	✓	✓	✓	✓
Chilli-Specific (Nai Miris) Model	✗	✗	✗	✓

1.3 Research Gap

A comprehensive review of existing literature reveals a significant gap in the development of integrated smart agriculture systems.

Firstly, there is a lack of synergy between AI-based computer vision systems and IoT-based environmental monitoring systems. Most research efforts treat these technologies independently, resulting in fragmented solutions that fail to provide holistic insights into crop growth A comprehensive review of existing literature reveals a significant gap in the development of integrated smart agriculture systems.

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Secondly, existing studies predominantly focus on broad classification of plant growth stages, rather than detailed, week-by-week phenological tracking. This limitation is particularly critical during early growth stages, where timely intervention can significantly impact yield [21].

Thirdly, many smart agriculture systems are designed as generalized frameworks, without considering crop-specific biological requirements. This reduces their effectiveness when applied to specialized crops such as Capsicum chinense, which have unique growth patterns and environmental needs [13].

Finally, current systems are largely reactive, focusing on detecting problems after they occur. There is a clear need for proactive systems that monitor crop development continuously and provide actionable guidance to prevent stress conditions before they arise [19].

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2. OBJECTIVE

2.1 Main Objective

To conceptualize, engineer, and validate a highly integrated IoT and Edge-AI-based crop growth monitoring system that continuously evaluates the weekly phenological development of *Nai Miris* (Capsicum chinense) against real-time micro-environmental sensor data, thereby providing smallholder farmers with proactive, stage-specific cultivation guidance to maximize crop yield and optimize resource utilization.

2.2 Specific Objectives

To systematically achieve the overarching main objective, the following specific, measurable, and highly technical sub-objectives have been established:

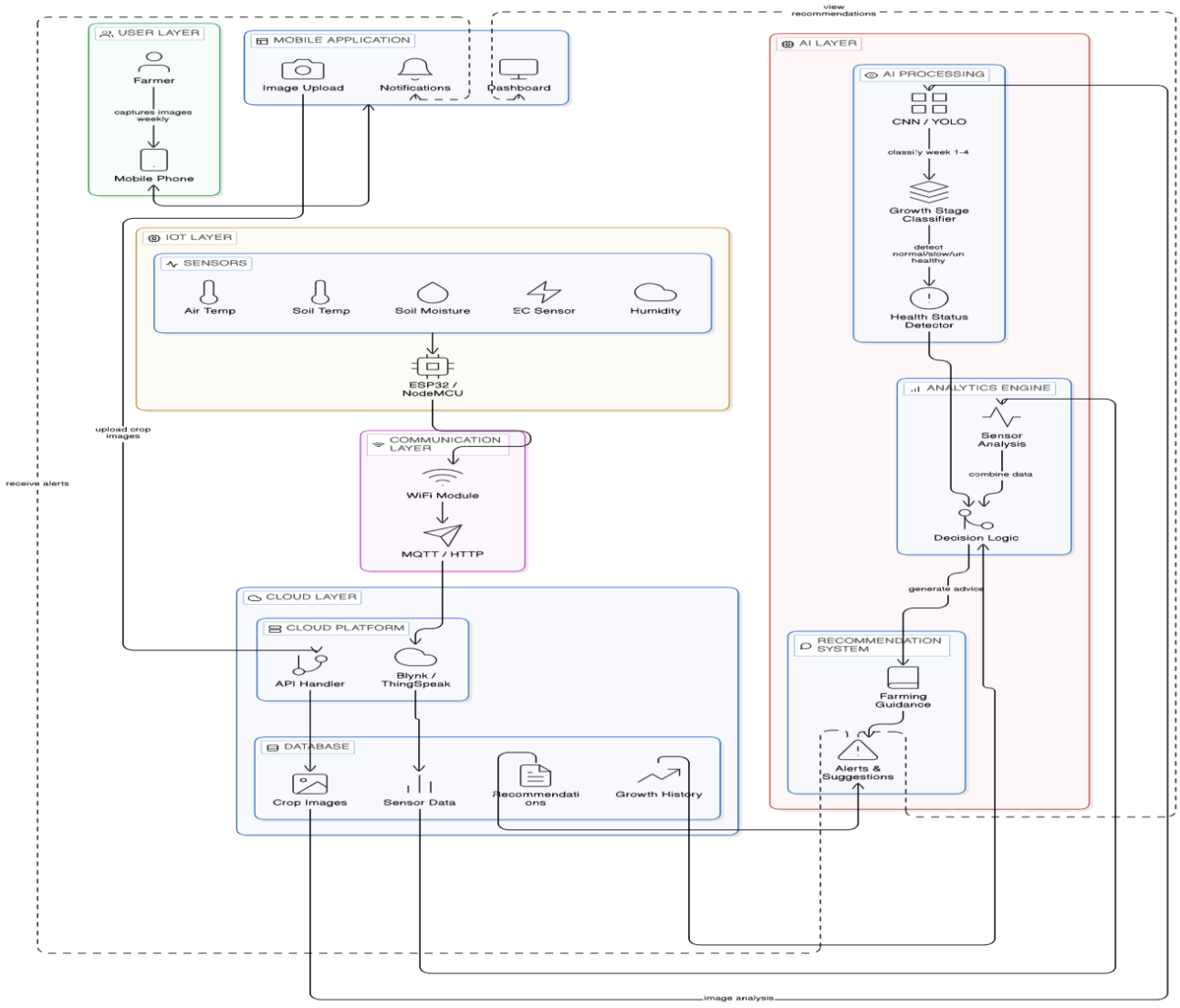
1. **Engineer an Edge-AI Computer Vision Pipeline:** To compile a proprietary, annotated dataset of Nai Miris phenology and subsequently train, optimize, and deploy lightweight deep learning architectures (specifically YOLOv8n for localization and ResNet-18 for classification) capable of accurately identifying the weekly growth stages (Weeks 1 through 4) using standard smartphone imagery.
2. **Develop a Robust IoT Environmental Sensing Network:** To design and assemble a low-cost, ESP32-microcontroller-based sensor node that reliably captures, digitally filters, and transmits real-time root-zone telemetry—including soil volumetric water content, electrical conductivity (EC), ambient temperature, and relative humidity—to a centralized cloud architecture via the MQTT protocol.
3. **Formulate a Multi-Variate Data Fusion and Diagnostic Algorithm:** To develop a cloud-hosted, rule-based inference engine that mathematically cross-references the AI-derived phenotypic growth state with the concurrent IoT environmental time-series data to definitively isolate the root causes of morphological anomalies (e.g., algorithmically distinguishing between drought stress and macronutrient lockout).
4. **Design an Interactive Digital Guidance Dashboard:** To architect a highly intuitive, user-centric mobile application that seamlessly visualizes real-time environmental analytics and delivers automated, actionable agronomic recommendations (e.g., "Increase irrigation frequency," "Apply high-nitrogen localized fertilizer") dynamically tailored to the specific developmental week of the monitored crop.

Empirically Validate System Efficacy and Computational Scalability: To conduct rigorous field trials in semi-controlled environments, quantitatively measuring the end-to-end network latency, the AI model's precision and recall (mAP and F1-score), and the empirical, comparative impact of the system's guidance on the crop's vegetative biomass and time-to-flowering against traditional manual cultivation methods.

3. METHODOLOGY

To ensure scientific rigor and practical applicability, this research will employ a formal Design Science Research (DSR) methodology. DSR is highly suited for Information Systems and software engineering research aimed at creating, deploying, and evaluating innovative IT artifacts intended to solve identified organizational or real-world operational problems. The methodology is systematically structured into four primary phases: dataset acquisition, hardware engineering, algorithm development, and empirical system validation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



The architectural framework of the proposed system is engineered as a highly distributed, multi-tier IoT and Edge AI topology. To ensure maximum scalability, fault tolerance, and low-latency processing, the architecture is structurally decoupled into four distinct, interacting layers: the Perception Layer, the Edge/Communication Layer, the Cloud Processing Layer, and the Application Layer.

Overview of Proposed Solution and Components

1. Perception Layer (Data Acquisition)

This foundational layer operates directly within the physical, harsh agricultural environment of the field or greenhouse. It consists of two highly distinct, parallel data acquisition streams:

- **The Telemetry Node:** A physical array of robust hardware sensors buried within the active root zone and mounted directly above the plant canopy. This array includes the analog capacitive soil moisture sensor, the vital EC probe, and the digital DHT22 module for ambient microclimate monitoring.
- **The Visual Node:** Unlike traditional systems that mount expensive, static field cameras, this architecture leverages the farmer's existing smartphone camera. The smartphone acts as an ad-hoc visual sensor, capturing high-resolution RGB images of the Nai Miris crop during the farmer's routine weekly physical inspection of the field.

2. Edge and Communication Layer

- **Microcontroller Gateway (ESP32):** The ESP32 microcontroller functions as the localized edge gateway for all environmental telemetry. It performs analog-to-digital conversion, digitally filters signal noise and erratic voltage spikes from the sensors, and packages the verified data points into optimized JSON strings.
- **Dual-Protocol Communication:** Recognizing the severe network instability that plagues open-field agriculture in developing regions, the system employs a highly robust dual-protocol design. Continuous, low-bandwidth telemetry data is pushed to the cloud using the MQTT protocol. MQTT's publish/subscribe model and tiny header size make it ideal for fluctuating 3G/4G connectivity. Conversely, the smartphone application transmits the heavy, high-resolution crop images to the cloud inference engine utilizing secure HTTP/REST APIs, ensuring data integrity during large file transfers.[7]

3. Cloud Processing and Analytics Layer

This layer serves as the cognitive, analytical core of the entire system, hosted on a highly scalable, distributed cloud provider (e.g., AWS or Google Cloud).

- **MQTT Broker & Time-Series Database:** Rapidly ingests, timestamps, and permanently stores the high-frequency sensor payloads, allowing for the generation of historical trend lines and predictive analytics.
- **AI Inference Engine:** When a new crop image is received via the mobile application's HTTP POST request, it is immediately routed to the containerized inference engine. The YOLOv8n model executes first, rapidly cropping the image to the exact region of interest (the plant canopy). This isolated, localized image is then passed through the ResNet-18

classification model, which performs a rapid matrix multiplication to output a normalized probability distribution across the four weekly growth stages.

- **Diagnostic Rule Engine (Data Fusion):** This critical microservice executes the core data fusion logic. Upon receiving a newly classified growth stage from the AI engine, it queries the time-series database for the most recent, concurrent environmental telemetry associated with that specific plant's zone. It mathematically compares the biological age of the plant to its visual stage, evaluates the soil parameters against hardcoded optimal thresholds for Nai Miris , and algorithmically generates the prescribed agronomic action.

4. Application Layer (User Interface)

A cross-platform mobile application, developed utilizing modern frameworks such as Flutter or React Native, serves as the primary, interactive touchpoint for the farmer. It is designed to abstract complex data into simple visuals, displaying real-time gauge charts for soil parameters, historical environmental trend graphs, the results of the weekly AI visual analysis, and critical push notifications containing the plain-text farming guidance.

The system operates on two distinct, concurrent data flows to maximize efficiency:

1. **Continuous Telemetry Flow (Passive):** Physical Sensors $\rightarrow$ ESP32 ADC Conversion $\rightarrow$ Wi-Fi transmission via MQTT $\rightarrow$ Cloud MQTT Broker $\rightarrow$ Time-Series Database $\rightarrow$ Live Mobile Dashboard Update.
2. **Event-Driven Diagnostic Flow (Active):** Farmer captures and submits crop image via App $\rightarrow$ Image uploaded via secure HTTP POST $\rightarrow$ Cloud AI Inference Engine extracts deep features $\rightarrow$ Model predicts Weekly Growth Stage $\rightarrow$ Diagnostic Engine queries recent IoT data from database $\rightarrow$ Engine synthesizes visual and sensor data via logic matrix $\rightarrow$ Custom Guidance Report generated $\rightarrow$ Urgent Push Notification dispatched Farmer's App.

Scalability and Security Considerations

- **Scalability:** The utilization of an MQTT broker architecture inherently supports thousands of concurrent, lightweight TCP connections. If a commercial farm expands operations from a 1-acre plot to a 10-acre plantation, the farmer simply procures and deploys additional, pre-configured ESP32 nodes without altering the core software architecture. Furthermore, the cloud infrastructure can utilize auto-scaling container orchestration (like Kubernetes) to seamlessly spin up additional AI inference instances to handle sudden spikes in image upload traffic during peak weekend inspection times.
- **Security:** Agricultural data integrity is paramount; unauthorized manipulation of telemetry could trigger incorrect fertilizer guidance, destroying a crop. To prevent this, all MQTT traffic between the edge nodes and the broker will be heavily encrypted using TLS/SSL cryptographic protocols. Furthermore, mobile image uploads and user authentication will strictly utilize secure, token-based authentication mechanisms (OAuth 2.0) to ensure absolute data privacy and integrity

5. USER REQUIREMENTS

Stakeholder Identification

The successful deployment and commercial viability of this system depend heavily on satisfying the distinct needs of three primary stakeholder groups:

1. **Smallholder and Commercial Farmers:** The primary end-users of the system. They require an intuitive, low-cost, and highly reliable decision-support tool to definitively improve Nai Miris crop yields, mitigate the risk of catastrophic crop failure, and drastically reduce the exorbitant costs associated with wasted fertilizer and agricultural inputs.
2. **Agricultural Extension Officers (DOA):** Secondary, macro-level users. With appropriate permissions, these government officers can utilize the aggregated, anonymized cloud data to accurately monitor regional crop health, predict localized harvest volumes, and disseminate broader, data-backed policy advice or emergency drought warnings.
3. **Agri-Tech Service Providers and Technicians:** Tertiary stakeholders comprising the entities that will manufacture, deploy, calibrate, and physically maintain the hardware infrastructure in the field as a scalable B2B service.

5.2 Functional Requirements

Table 1 outlines the specific, measurable behaviors and operational functions the integrated system must execute to satisfy the core objectives of the research.

Table 2 functional Requirement

Req ID	**Component**	**Requirement Description**
FR1	Sensor Node	The physical ESP32 system must accurately read soil moisture, EC, temperature, and humidity from analog/digital pins at predefined intervals (e.g., every 15 minutes).

FR2	Telemetry	The ESP32 microcontroller must successfully serialize and transmit sensor payloads to the centralized cloud broker via the MQTT protocol over a local Wi-Fi connection.
FR3	Image Capture	The mobile application interface must allow the user to activate the device camera, capture a clear RGB image of the target plant, and upload it to the cloud.
FR4	AI Inference	The cloud-hosted backend must accurately process the uploaded image using the deployed YOLOv8 and ResNet-18 models to output a mathematically predicted growth stage (Week 1, 2, 3, or 4).
FR5	Data Fusion	The algorithmic engine must mathematically compare the AI-predicted phenotypic stage with the plant's recorded chronological age to definitively detect developmental delays.
FR6	Diagnostics	If a developmental delay is flagged, the system must

		immediately query current IoT time-series data to determine if specific root-zone parameters (e.g., EC) are falling outside the established optimal threshold for Nai Miris.
FR7	Guidance	The system must automatically generate human-readable, highly actionable agronomic recommendations (e.g., "Increase nitrogen fertilizer application") based strictly on the fusion of the AI and IoT data matrices.
FR8	Dashboard	The mobile application must intuitively visualize real-time IoT data streams utilizing dynamic gauge charts, historical line graphs, and clear status indicators.
FR9	Alerts	The system architecture must trigger instantaneous push notifications to the user's mobile device immediately upon detecting critical abnormal environmental conditions or generating a new weekly guidance report.

5.3 Non-Functional Requirements

Table 2 outlines the stringent criteria utilized to judge the operation, stability, and security of the proposed system.

Table 3 Non-functional Requirement

Req ID	**Category**	**Requirement Description**
NFR1	Performance	The entire pipeline—from image upload, AI inference, and diagnostic rule processing—must complete execution and return a rendered result to the mobile application within a maximum latency threshold of 3.0 seconds.
NFR2	Reliability	The overarching cloud backend, database, and critical MQTT broker must be engineered to maintain a high-availability uptime metric of 99.9%.
NFR3	Usability	The mobile application user interface must be designed to be highly intuitive, utilizing clear visual iconography and supporting native local languages (Sinhala/Tamil) to seamlessly accommodate rural users with limited digital literacy.

NFR4	Power	The ESP32 sensor edge node must be programmatically optimized for extreme low power consumption (utilizing deep sleep cycles), potentially supporting integration with 5V small-scale solar panels for continuous, off-grid open-field operation.
NFR5	Security	All data transmissions between the localized edge nodes, the mobile application, and the cloud servers must be rigorously encrypted utilizing industry-standard TLS/SSL cryptographic protocols to prevent unauthorized interception or data manipulation.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for this technology consists of medium-to-large-scale chili farmers, specialized agricultural cooperatives, and commercial plantation companies operating primarily within Sri Lanka's dry and intermediate climatic zones (such as Anuradhapura, Puttalam, Matale, and Kurunegala), where the vast majority of commercial condiment crops are cultivated.The idealized customer persona is a progressively minded agribusiness manager or a younger, tech-adoptive farmer who currently cultivates high-value hybrid crops (like Nai

Miris). This persona is highly motivated to adopt technology due to profound frustration with unpredictable seasonal yields, the devastating impact of sudden droughts, and the rapidly escalating, prohibitive costs of imported chemical fertilizers and manual labor A highly lucrative secondary market also exists in the rapidly expanding sector of urban agriculture, specifically poly-tunnel or net-house cultivators , who absolutely require precise, 24/7 environmental control to justify their high initial infrastructure investments.

Value Proposition

Current agricultural inputs in Sri Lanka—particularly imported fertilizers, pesticides, and hired physical labor—are exorbitantly expensive. Economic surveys indicate that manual labor costs alone represent a staggering 64% to 75% of total cultivation expenses for various chilies and onions in Sri Lanka.Our integrated system offers a definitive **"Precision-Yield Guarantee."** By shifting the agricultural paradigm from a reactive, damage-control model to a proactive, AI-guided precision model, farmers can empirically prevent yield degradation weeks before visual damage becomes severe. This methodology maximizes the harvest output of high-value Nai Miris (which consistently fetches premium wholesale market prices) while simultaneously, and critically, reducing the wasted application of expensive, unnecessary fertilizers and irrigation water through highly targeted, data-driven dosing.

Revenue Model

To ensure rapid market penetration and sustainable recurring revenue, the commercialization strategy will employ a modern, hybrid **Hardware-as-a-Service (HaaS) / Freemium Software Model**:

1. **Hardware Sales (Cost Recovery + Margin):** The physical IoT sensor kit—comprising the pre-programmed ESP32, the sensor array, and durable, weatherproof casing—is sold to the farmer at a fixed, highly affordable margin. This establishes the necessary physical hardware footprint on the farm without posing a massive barrier to entry.
2. **Freemium Software Access (SaaS):**
    - *Basic Tier (Free for Life):* Allows farmers to utilize the mobile app to view basic, real-time IoT telemetry (ambient temperature, simple soil moisture) without access to the advanced AI analysis or data fusion engine. This builds daily active users (DAU) and brand trust.
    - *Premium Tier (Subscription - Estimated LKR 1000/month per acre):* Unlocks the full power of the platform: weekly AI growth stage analysis, advanced EC nutrient tracking, cross-referenced data fusion diagnostics, and the automated, stage-based prescriptive farming guidance.

Cost Estimation and Pricing Strategy

The hardware Cost of Goods Sold (COGS) has been meticulously engineered to be highly disruptive to existing market alternatives. By utilizing readily available, mass-produced, off-the-shelf micro-components, costs are kept exceptionally low. For example, local Sri Lankan retail pricing for the highly accurate DHT22 temperature and humidity module is approximately LKR 1200 to LKR 1280. Capacitive soil moisture sensors can be procured

locally for under LKR 300.Consequently, the entire edge sensor node can be assembled, soldered, and flashed with firmware for under LKR 6000. The complete kit will be retail-priced at an introductory rate of LKR 10,000. This ensures a sustainable gross profit margin on the hardware while remaining vastly more accessible than imported, enterprise-grade agricultural telemetry systems that frequently cost hundreds of dollars per node. Software infrastructure costs (cloud hosting, database reads/writes) are highly elastic, scaling linearly with user acquisition, and are easily covered by the recurring revenue generated from the premium subscription tier.

Competitive Advantage

The Sri Lankan Agri-Tech landscape currently contains several ambitious competitors, most notably platforms such as GeoGoviya, Govi Sahaya, and SenzAgro.However, our proposed system possesses a distinct, highly defensible competitive moat.

- **Against GeoGoviya:** While GeoGoviya provides excellent macro-climate data and regional drought warnings, it fundamentally lacks field-level, plant-specific micro-monitoring; it cannot tell a farmer that the EC in row 4 is too low for week 2 of growth.
- **Against Govi Sahaya:** Govi Sahaya's embedded AI is focused primarily on diagnosing existing pathologies (diseases and pests).It does not track continuous, healthy phenological progression. **The Ultimate Advantage:** We are introducing the only localized platform offering **integrated, crop-specific (Nai Miris), weekly phenological progression tracking intrinsically combined with live, root-zone soil telemetry**. This continuous, dual-stream data ingestion creates a massive, proprietary dataset that continuously improves the AI's predictive accuracy over time, creating a formidable technological barrier to entry for any new market competitors.

7. BUDGET AND JUSTIFICATION

The proposed budget detailed below ensures the adequate procurement of necessary hardware components for rapid prototyping, robust cloud infrastructure for backend deployment and API hosting, and requisite software licenses for mobile application distribution.

Table 3: Minimum Viable Product (MVP) Cost Estimation and Hardware Justification.

Table 4 Minimum Viable Product (MVP) Cost Estimation and Hardware Justification.

Expense Category	Item / Component Description	Estimated Cost (LKR)	Technical Justification
Hardware	ESP32 Microcontrollers (x3)	7,500	Core processing and Wi-Fi transmission units required to build multiple test nodes for spatial field coverage.[6]
Hardware	DHT22 Temp/Humidity Sensors (x3)	3,600	Selected for highly accurate, digital ambient microclimate monitoring above the canopy, superior to the cheaper DHT11.[34]
Hardware	Capacitive Soil Moisture Sensors (x3)	750	Selected for durable, anti-corrosive measurement of root-zone volumetric water

			content, vastly outlasting standard resistive forks.[33]
Hardware	Analog EC Sensors & Probes (x3)	15,000	The most critical component for precisely measuring macronutrient salt availability in the soil solution.[6]
Hardware	Power supplies, wiring, IP67 waterproof housing	5,000	Absolutely necessary to protect delicate microelectronics during field deployment in harsh, humid, and rainy agricultural environments.
Software	Cloud Hosting (AWS/GCP) - 6 months	30,000	Essential for hosting the MQTT message broker, the high-frequency time-series database, and executing the heavy AI inference engine APIs.
Software	Apple Developer / Google Play Registration	15,000	Mandatory registration fees required for officially deploying the compiled mobile application to end-users for field testing.

Misc / Travel	Travel to agricultural testing sites	10,000	Required for continuous field testing, physical dataset image collection, and empirical agronomic validation in the dry zones.
Total		**86,850 LKR**	

Note: The cost of human effort, software development, and deep learning model training is considered zero for the duration of this specific academic research project, as all rigorous development and engineering will be undertaken exclusively by the student researcher.

8. WORK BREAKDOWN STRUCTURE (WBS)

To ensure the timely, systematic, and rigorous completion of the project within the academic timeframe, the complex overall workload is divided into five primary, sequential phases, with clear deliverables associated with each step.

Table 4: Work Breakdown Structure (WBS) Task Allocation

Table 5 Work Breakdown Structure (WBS) Task Allocation

Phase	Task Name	Detailed Description	Final Deliverable
Phase 1	Project Initiation & Literature Review	Conduct an exhaustive, systematic study of precision agriculture, deep	Finalized Project Proposal, Literature

		learning vision models, IoT telematics frameworks, and specific Nai Miris agronomy.	Review, and System Architecture Blueprint.
Phase 2	Data Collection & AI Model Engineering	Cultivate a control group of Nai Miris, capture weekly high-resolution images, annotate via Roboflow, and train YOLOv8n and ResNet-18 models.	Trained, optimized, and mathematically quantized Edge-AI models (TFLite format).
Phase 3	IoT Hardware Assembly & Integration	Physically solder and assemble ESP32 nodes, mathematically calibrate DHT22, EC, and moisture sensors, and establish a secure MQTT cloud connection.	A fully functional, weather-proof telemetry transmission hardware system.
Phase 4	Cloud Backend & Data Fusion Logic	Develop the cloud inference REST API and program the complex diagnostic rule engine that mathematically fuses the IoT telemetry with the AI vision classifications.	A functional, cloud-hosted algorithmic backend capable of autonomously generating guidance.

Phase 5	Mobile App Dev & Empirical Validation	Build the cross-platform user dashboard. Deploy the entire integrated system in a physical test field to measure accuracy, latency, and agronomic impact.	The final integrated system, deployed Mobile App, and the comprehensive Final Dissertation document.

9. GANTT CHART

The following schedule outlines the strict timeline for the project execution across a standard 8-month academic calendar, explicitly tracking milestones aligned with an undergraduate final-year research project lifecycle.[37]

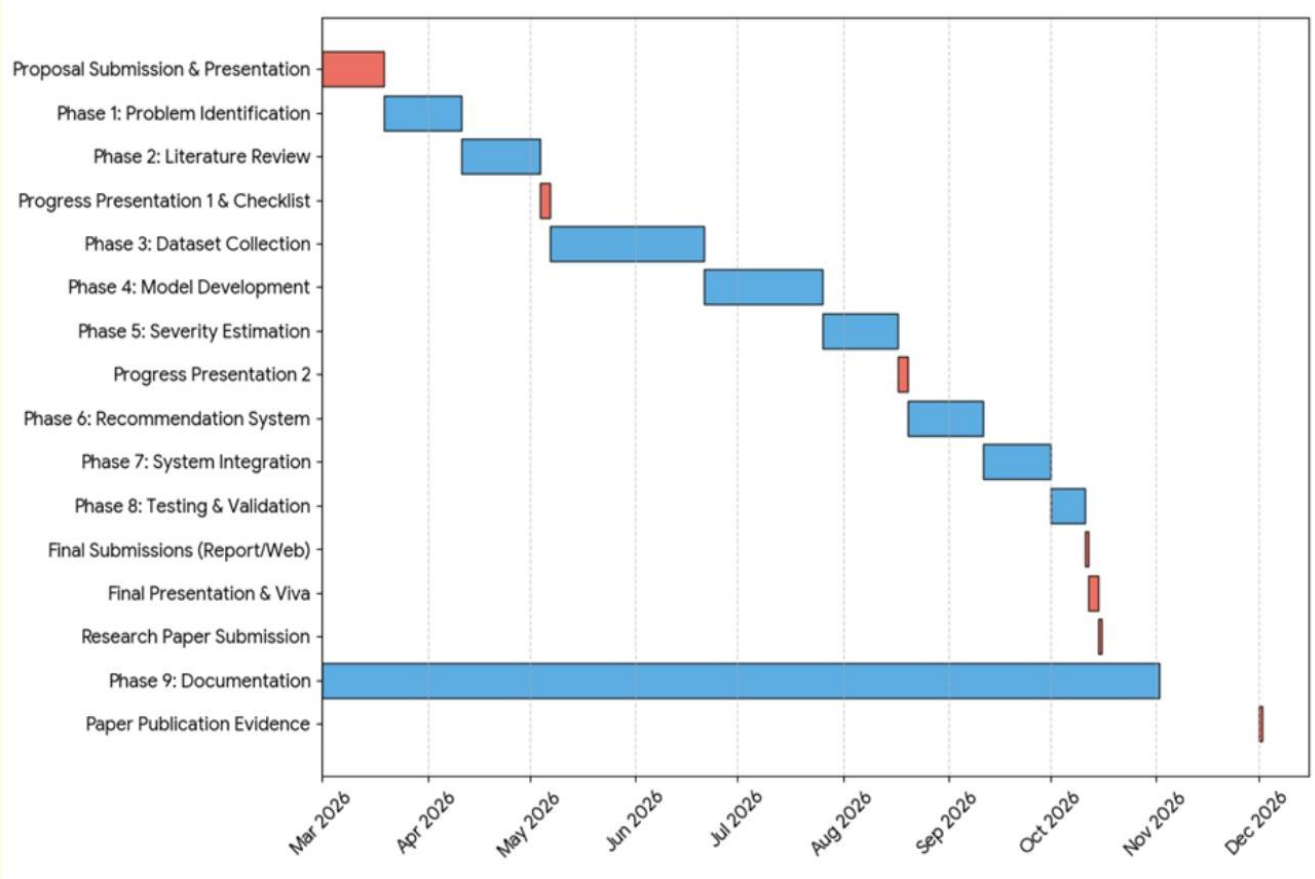


Figure 2 Gaant chart

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11.APPENDICES

Appendix A: Expected Weekly Growth Phenology for Nai Miris Cultivation

The following biological parameters are derived from standard agronomic guidelines and dictate the specific physical traits the deep learning models are trained to detect during the critical one-month early vegetative phase.

Table 6: Expected Weekly Growth Phenology for Nai Miris Cultivation

Table 6 Expected Weekly Growth Phenology for Nai Miris Cultivation

Development Stage	**Biological Timeline**	**Primary Phenotypic Characteristics to be Detected by AI Vision Models**
Stage 1	Week 1	**Early vegetative:** Emergence from soil. Cotyledons (seed leaves) clearly present. Emergence of the very first set of true leaves. Exceptionally small overall canopy radius.
Stage 2	Week 2	**Leaf development:** Rapid spatial expansion of true leaves. The primary stem begins to visibly thicken and darken. Cotyledons may begin to naturally senesce (wither).
Stage 3	Week 3	**Advanced vegetative:** Significant, measurable increase in leaf density and overall canopy width. Primary

		lateral branching initiates at the lower nodes.
Stage 4	Week 4	**Pre-flowering:** Highly complex, multi-tiered branching structure established. The appearance of initial, microscopic floral buds at the upper internodes.

Appendix B: Sensor Calibration Mathematical Formulas

For true precision agriculture, raw analog electrical readings from the field sensors must be mathematically calibrated into standardized physical units before the data fusion engine can process them.

- **Electrical Conductivity (EC) Temperature Compensation:**
  Raw electrical conductivity readings in the soil are highly sensitive to microscopic temperature fluctuations. To ensure the diagnostic engine does not trigger a false nutrient deficiency warning due to a drop in ambient temperature, the system will utilize the accompanying digital DHT22 data to mathematically normalize the raw EC reading to a standard reference temperature of 25°C using the following LaTeX formula:

$$EC_{25} = \frac{EC_t}{1 + \alpha(T - 25)}$$

  Where:
  - EC_{25} represents the standardized, temperature-compensated Electrical Conductivity.
  - EC_t represents the raw, uncalibrated analog reading captured at the current soil temperature T.
  - α represents the specific temperature compensation coefficient (typically hardcoded at 0.019 °C^{-1} for standard agricultural saline solutions).